

Conducting Forensic Investigations on System Memory (4e)

Digital Forensics, Investigation, and Response, Fourth Edition - Lab 10

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Time on Task:

5 hours, 0 minutes

Progress:

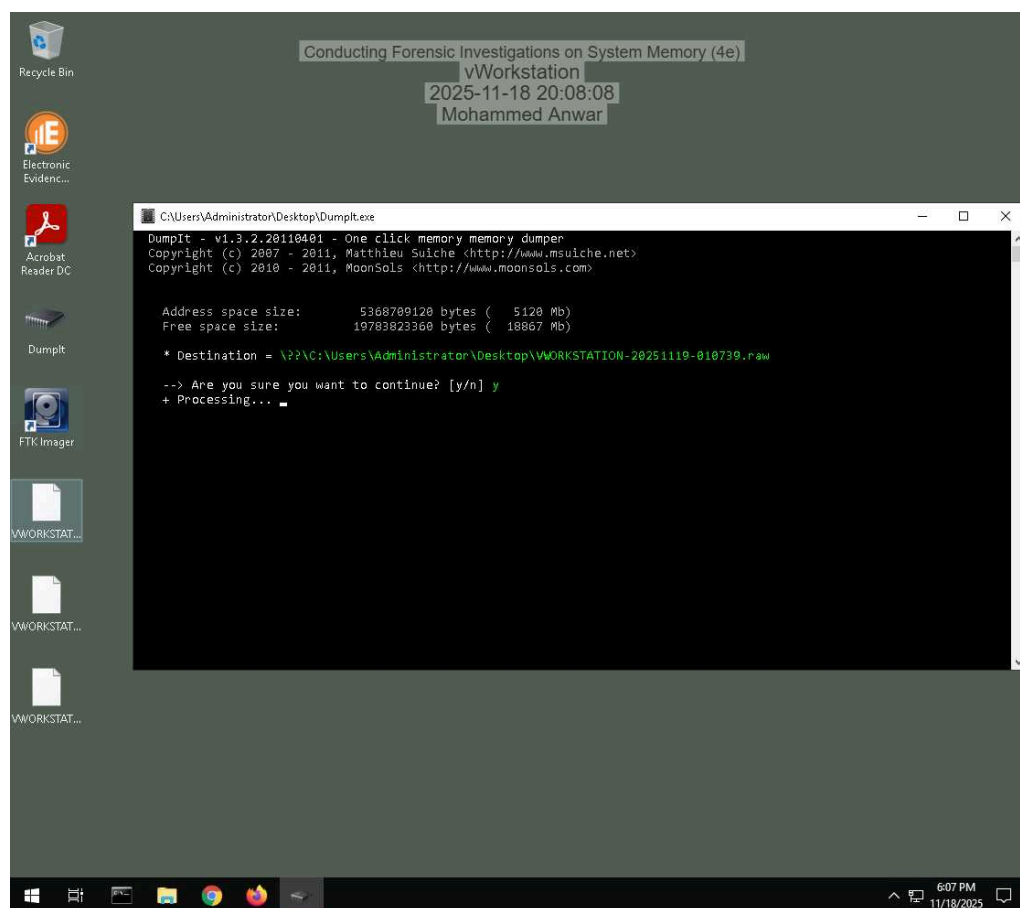
94%

Report Generated: Tuesday, November 18, 2025 at 10:33 PM

Section 1: Hands-On Demonstration

Part 1: Capture Memory using DumpIt

3. Make a screen capture showing the **Dumplt success notification**.

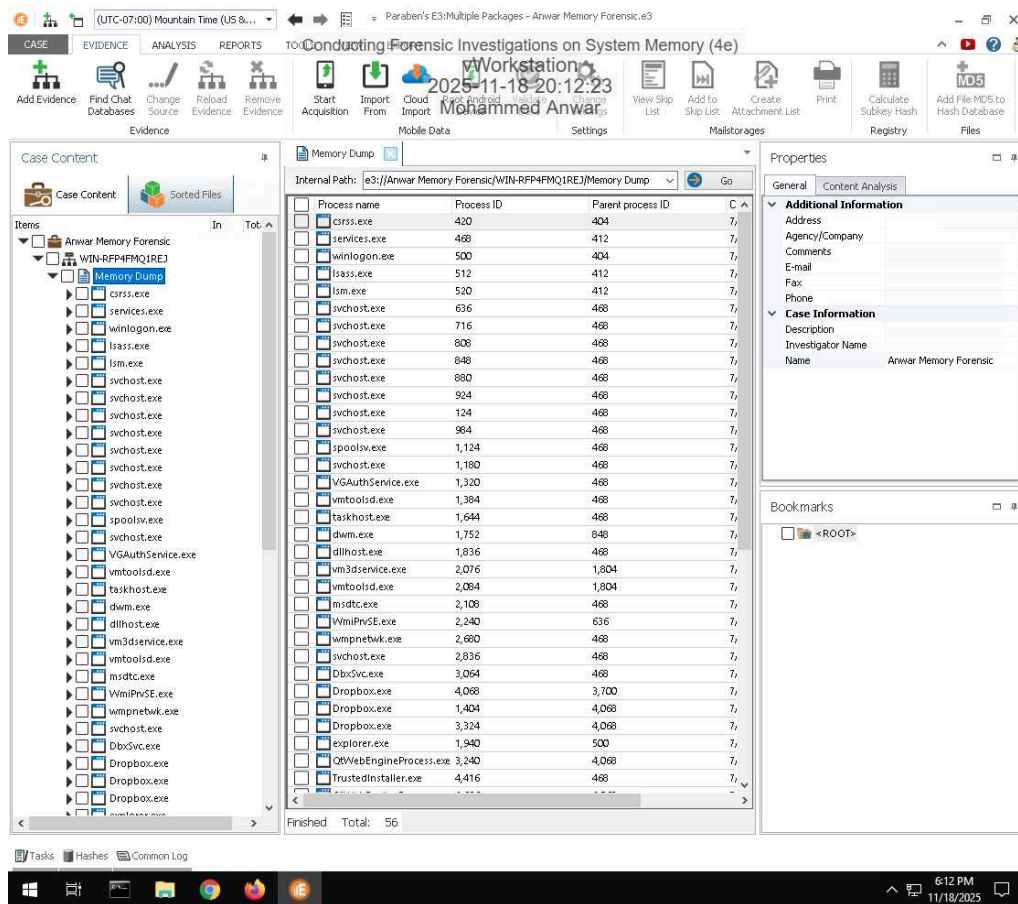


Part 2: Analyze Memory using E3

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8. Make a screen capture showing the list of processes in the memory dump.



10. Record the start times for the oldest process and the newest process.

4:24;52 AM - oldest

15. Document your findings for the conhost.exe process. What is it and what is it used for?

windows process that helps manage console windows on computer

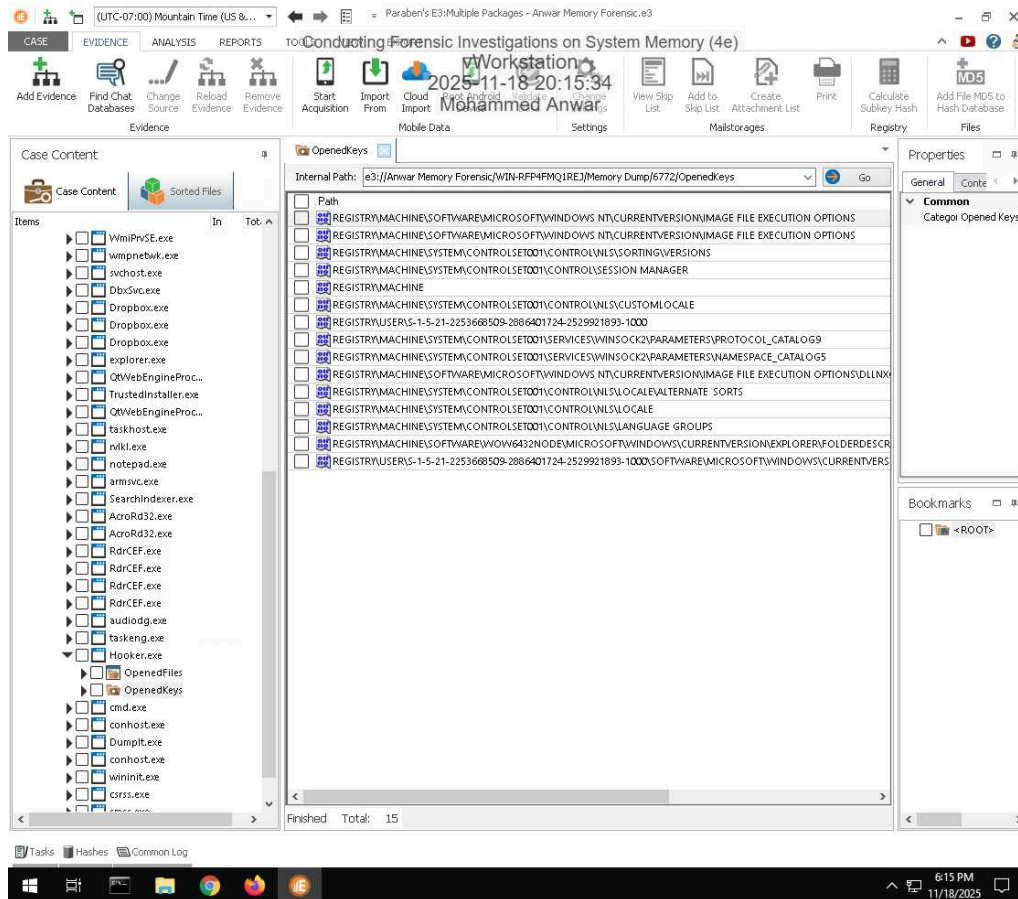
17. Document your findings for the hooker.exe process. What is it and what is it used for?

used for recording keyboard and mouse inputs

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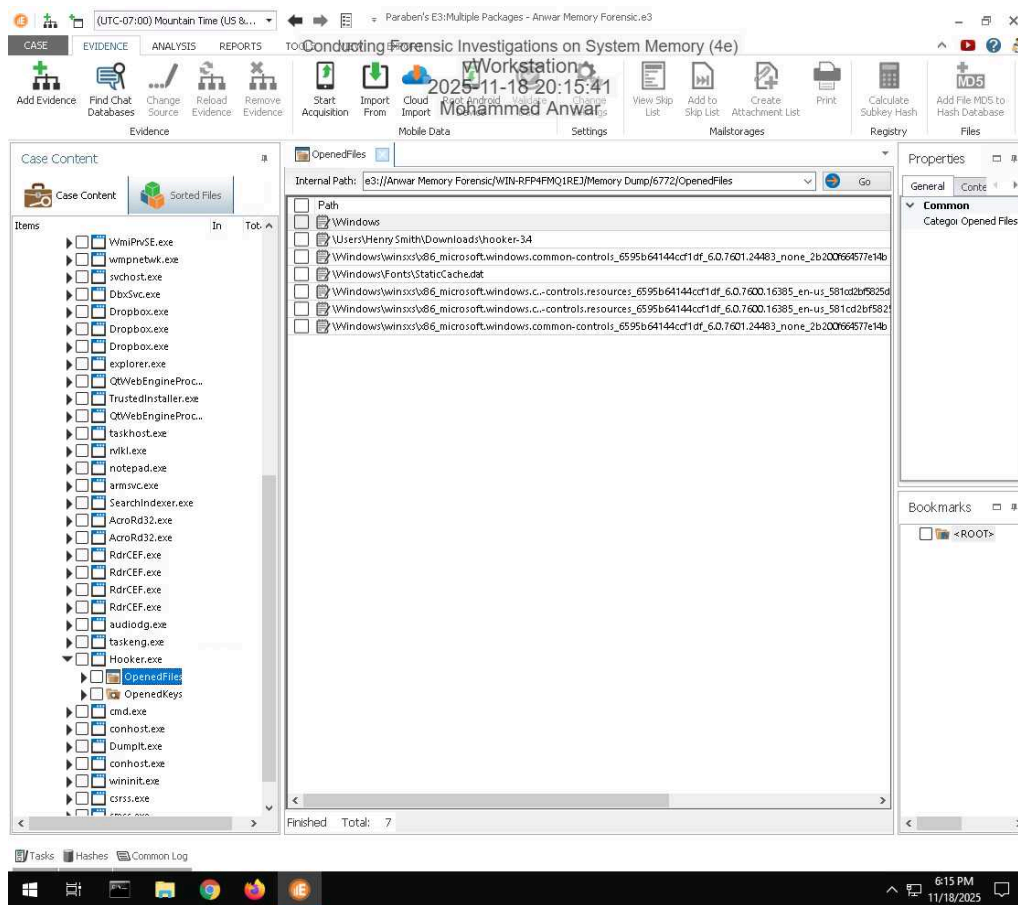
21. Make a screen capture showing the registry keys opened by the Hooker.exe process.



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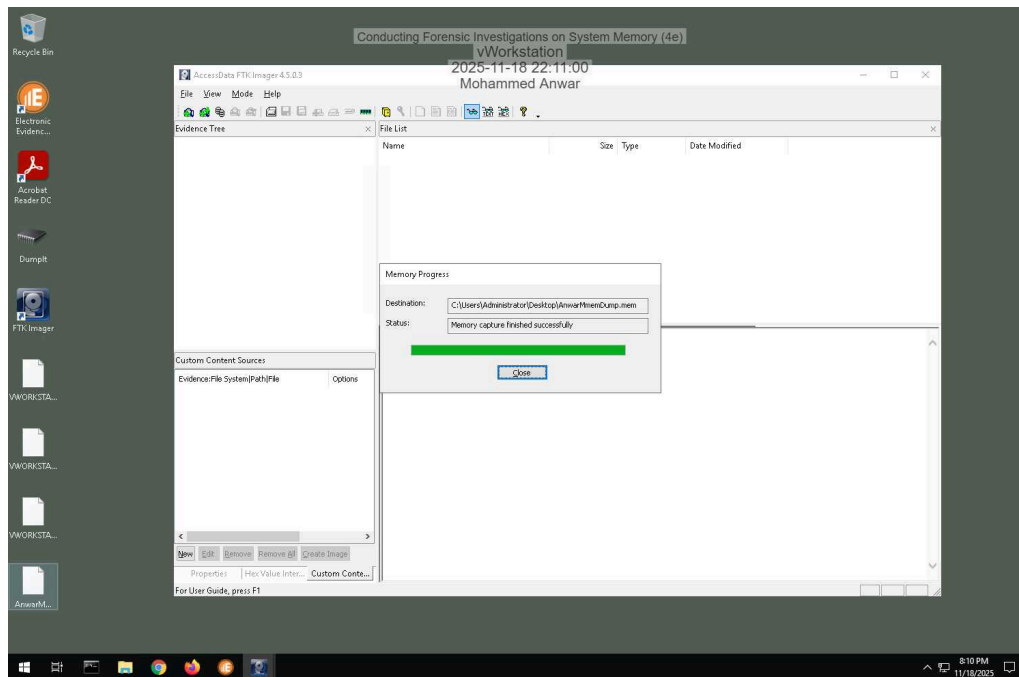
23. Make a screen capture showing the files opened by the hooker.exe process.



Section 2: Applied Learning

Part 1: Capture Memory using FTK Imager

6. Make a screen capture showing the *Memory capture finished successfully* confirmation.



Part 2: Analyze Memory using Volatility

7. **Document** your findings for the rvkl.exe process. What is it and what is it used for?

Main executable for the revealer keylogger software

9. **Document** whether any processes are flagged as hidden.

evil.exe

12. **Document** whether the netscan module displays network usage associated with the Hooker.exe or rvkl.exe processes.

yes it does

15. **Document** any information you were able to gather about port 56610.

it is categorized as a client port

26. **Make a screen capture** showing the **DensityScout** results.

The screenshot shows a Windows Command Prompt window with the following content:

```
Administrator: Command Prompt

:44 UTC+0000
0xbe0308d0 TCPv4 0.0.0.0:49152 0.0.0.0:49152 LISTENING 808 svchost.exe
0xbe31c9e0 TCPv4 0.0.0.0:49153 0.0.0.0:49153 LISTENING 808 svchost.exe
0xbe31c9e0 TCPv6 :::49153 0.0.0.0:49153 LISTENING 808 svchost.exe
0xbe3ec410 TCPv4 0.0.0.0:135 0.0.0.0:135 LISTENING 716 svchost.exe
0xbe3ec410 TCPv6 :::135 0.0.0.0:135 LISTENING 716 svchost.exe
0xbe3ec540 TCPv4 0.0.0.0:135 0.0.0.0:135 LISTENING 716 svchost.exe
0xbe3f0ae0 TCPv4 0.0.0.0:49152 0.0.0.0:49152 LISTENING 412 wininit.exe
0xbe3fc520 TCPv4 0.0.0.0:49152 0.0.0.0:49152 LISTENING 412 wininit.exe
0xbe3fc520 TCPv6 :::49152 0.0.0.0:49152 LISTENING 412 wininit.exe
0xbe06c970 TCPv4 127.0.0.1:50305 127.0.0.1:50304 ESTABLISHED -1
0xbe4d3010 TCPv4 127.0.0.1:50308 127.0.0.1:50307 ESTABLISHED -1
0xbe811010 TCPv4 192.168.106.144:54929 162.125.19.130:443 ESTABLISHED -1
0xbeba5260 UDPv4 192.168.106.144:64439 *: * 896 chrome.exe 2021-07-12 12:34

:42 UTC+0000
0xbeef5150 UDPv4 0.0.0.0:55769 *: * 984 svchost.exe 2021-07-12 12:33

:32 UTC+0000
0xbeceda00 TCPv4 0.0.0.0:49154 0.0.0.0:49154 LISTENING 924 svchost.exe
0xbf37ad90 TCPv4 0.0.0.0:49154 0.0.0.0:49154 LISTENING 924 svchost.exe
0xbf37ad90 TCPv6 :::49154 0.0.0.0:49154 LISTENING 924 svchost.exe
0xbec08240 TCPv4 192.168.106.144:56654 255.255.255.255:443 CLOSED -1
0xbec3d2d0 TCPv4 192.168.106.144:56655 255.255.255.255:443 CLOSED -1
0xbfdc1d00 UDPv4 127.0.0.1:63650 *: * 2836 svchost.exe 2021-07-12 10:25

:03 UTC+0000
0xbf971b00 TCPv4 192.168.106.144:54934 255.255.255.255:443 CLOSED -1

C:\Users\Administrator>C:\volatility.exe -f "C:\Memory Forensics Evidence\Win-RFP4FMQ1REJ.bin" --profile=Win7SP1x64 filescan | findstr rv
lkl.exe
Volatility Foundation Volatility Framework 2.6
0x00000000b0dd2a130 15 0 R--r-d \Device\HarddiskVolume1\Windows\System32\rvlkl.exe

C:\Users\Administrator>C:\volatility.exe -f "C:\Memory Forensics Evidence\Win-RFP4FMQ1REJ.bin" --profile=Win7SP1x64 filescan | findstr ho
oker.exe
Volatility Foundation Volatility Framework 2.6

C:\Users\Administrator>C:\volatility.exe -f "C:\Memory Forensics Evidence\Win-RFP4FMQ1REJ.bin" --profile=Win7SP1x64 filescan | findstr ho
oker.exe
Volatility Foundation Volatility Framework 2.6

C:\Users\Administrator>C:\densityscout.exe -p 0.1 "C:\ExtractedFiles"

DensityScout (Build 45)
by Christian Wojner

C:\Users\Administrator>
```

Section 3: Challenge and Analysis

Part 1: Identify Malicious Connections

Document the three processes that connected to 205.134.253.10:4444.

rvlkl.exe hooker.exe svchost.exe

Document the name and purpose of the software you discovered.

revealer keylogger - surveillance software hooker.exe - credential stealing malware

Part 2: Identify Malicious Processes

Make a screen capture showing the fixtureComputer.exe process, and all those below it, in the pslist output.

```

C:\Users\Administrator>cmd.exe /c pslist
PID      Name
-----
408      services.exe
984      svchost.exe
808      svchost.exe
4336     SearchIndexer.exe
1404     Dropbox.exe
5444     taskeng.exe
1124     spoolsv.exe
5908     AcroRd32.exe
512      lsass.exe
716      svchost.exe
2108     msdtc.exe
4068     Dropbox.exe
6000     conhost.exe
1752     dwm.exe
5952     RdrCEF.exe
3240     QtWebEnginePro
1180     svchost.exe
1644     taskhost.exe
3064     DbxSvc.exe
6972     RdrCEF.exe
2208     RdrCEF.exe
2240     WmIPrvSE.exe
1940     explorer.exe
848      svchost.exe
636      svchost.exe
2144     armsvc.exe
2636     AcroRd32.exe
4096     QtWebEnginePro
2076     vmtoolsd.exe
6268     audiodg.exe
4224     rvlkl.exe
1384     vmtoolsd.exe
924      svchost.exe
124      svchost.exe
4720     taskhost.exe
412      wininit.exe
3324     Dropbox.exe
364      notepad.exe
6080     RdrCEF.exe
520      lsm.exe
6772     Hooker.exe
1836     dlhst.exe
1320     VSAuthService.exe
3268     cmd.exe
508      winlogon.exe
880      svchost.exe
2836     svchost.exe
2084     vmtoolsd.exe
  
```


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Make a screen capture showing the output of the yarascan.

Incomplete

Part 3: Identify Privilege Escalation

Make a screen capture showing the output of your privilege comparison.

```
Select Administrator: Command Prompt
C:\Users\Administrator>C:\Volatility.exe -f 'C:\Memory Forensics\Volatility\Win7SP1\FMQ1REJ.bin' --profile=Win7SP1x64 psxview
Volatility Foundation Volatility Framework 2.6
Offset(P)      Name          PID  psscan  psscan  thrdproc  pspcid  csrss  session  deskthrd  ExitTime
-----
0x00000000be320b00  services.exe  468  True   False   False    True   True   True     False
0x00000000be0eab00  svchost.exe   984  True   False   False    True   True   True     True
0x00000000be0eab00  svchost.exe   808  True   False   False    True   True   True     True
0x00000000bedfcd00  SearchIndexer.exe 4336 True   False   False    True   True   True     True
0x00000000bd5d7900  Dropbox.exe   1404 True   False   False    True   True   True     True
0x00000000bd1af000  taskeng.exe   6544 True   False   False    True   True   True     True
0x00000000be15ab00  spoolsv.exe   1124 True   False   False    True   True   True     True
0x00000000bd137000  AcroRd32.exe  5508 True   False   False    True   True   True     True
0x00000000be331000  lsass.exe     512  True   False   False    True   True   True     False
0x00000000be3a0400  svchost.exe   716  True   False   False    True   True   True     True
0x00000000bd8dc430  msdtc.exe     2108 True   False   False    True   True   True     True
0x00000000bd548790  Dropbox.exe   4068 True   False   False    True   True   True     True
0x00000000bd493000  conhost.exe   6000 True   False   False    True   True   True     True
0x00000000bdfa0600  dwm.exe       1752 True   False   False    True   True   True     True
0x00000000bd3fc000  RdrCEF.exe    5952 True   False   False    True   True   True     True
0x00000000bd5e6000  QtWebEnginePro 3240 True   False   False    True   True   True     True
0x00000000be1073b0  svchost.exe   1180 True   False   False    True   True   True     True
0x00000000bd710000  taskhost.exe  1644 True   False   False    True   True   True     True
0x00000000bd537000  DbxSvc.exe    3064 True   False   False    True   True   True     True
0x00000000bd3b2000  RdrCEF.exe    6072 True   False   False    True   True   True     True
0x00000000bccf1000  RdrCEF.exe    2308 True   False   False    True   True   True     True
0x00000000bd3d3700  WmiPrvSE.exe  2240 True   False   False    True   True   True     True
0x00000000bd48e0c0  explorer.exe  1940 True   False   False    True   True   True     True
0x00000000be02d060  svchost.exe   848  True   False   False    True   True   True     True
0x00000000be3b1060  svchost.exe   636  True   False   False    True   True   True     True
0x00000000bd157060  armsvc.exe    2144 True   False   False    True   True   True     True
0x00000000bd3205e0  AcroRd32.exe  2636 True   False   False    True   True   True     True
0x00000000bd768000  QtWebEnginePro 4696 True   False   False    True   True   True     True
0x00000000bdcd0eb00  vmtoolsd.exe  2876 True   False   False    True   True   True     True
0x00000000bd160000  audiodg.exe   6268 True   False   False    True   True   True     True
0x00000000bd160000  nvlvl.exe     4224 True   False   False    True   True   True     True
0x00000000bd0c0000  vmtoolsd.exe  1304 True   False   False    True   True   True     True
0x00000000be05b060  svchost.exe   924  True   False   False    True   True   True     True
0x00000000be05b000  svchost.exe   124  True   False   False    True   True   True     True
0x00000000be35d320  taskhost.exe  4720 True   False   False    True   True   True     True
0x00000000be2c5760  wininit.exe   412  True   False   False    True   True   True     True
0x00000000bd982b00  Dropbox.exe   3324 True   False   False    True   True   True     True
0x00000000bd8f0000  notepad.exe   364  True   False   False    True   True   True     True
0x00000000bf1e0600  RdrCEF.exe    6080 True   False   False    True   True   True     True
0x00000000be324b00  lsass.exe     520  True   False   False    True   True   True     False
0x00000000bcc1f060  Hooker.exe    6772 True   False   False    True   True   True     True
0x00000000bd085000  dillhost.exe  1836 True   False   False    True   True   True     True
0x00000000bd060000  VSAUTHService.exe 1320 True   False   False    True   True   True     True
0x00000000be0fc030  cmd.exe       3268 True   False   False    True   True   True     True
0x00000000be32e5f0  winlogon.exe  508  True   False   False    True   True   True     True
0x00000000be046000  svchost.exe   880  True   False   False    True   True   True     True
0x00000000bd0e1060  svchost.exe   2836 True   False   False    True   True   True     True
0x00000000bd469b00  vmtoolsd.exe  2084 True   False   False    True   True   True     True
```